

David Tong Standard Model Solutions — Sheet 4: The Standard Model

D. Tong, The Standard Model, Example Sheet 4; 5 worked problems.

Yucheng (Luca) Jin

Quantum Field Theory

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These prompts paraphrase David Tong's The Standard Model, Example Sheet 4 (March 2026). Problem numbers, starred status, equations, and guidance follow the official sheet.

1. Sheet 4, Problem 1: Running Gauge Couplings

- a) For $SU(N_c)$ with N_f massless fundamental Dirac fermions and N_s massless fundamental scalars, use

$$\frac{1}{g^2(\mu)} = \frac{1}{g_0^2} - \frac{11N_c - 2N_f - N_s/2}{3(4\pi)^2} \log \frac{\Lambda_{UV}^2}{\mu^2}. \quad (1.1)$$

- i) Prove that the Standard Model $SU(2)$ sector is asymptotically free.
- ii) Without a Higgs boson, estimate the strong-coupling scale Λ_{weak} from $\alpha_W(100 \text{ GeV}) \simeq 1/30$.
- iii) If the Higgs did not condense, explain the effect on the weak force of a QCD condensate $\langle \bar{q}_{Li} q_{Rj} \rangle \simeq -\Lambda_{\text{QCD}}^3 \delta_{ij}$.
- iv) Given $\alpha_s(100 \text{ GeV}) \simeq 0.1$, estimate the scale $\mu \gg 90 \text{ GeV}$ where $\alpha_s = \alpha_W$.

Solution.

- i) For Standard Model $SU(2)$, we have $N_c = 2, N_s = 1$ and $N_f = 3 \cdot (3 + 1)/2$, where the 3 comes from three generations and the 4 comes from 3 colour of quarks and leptons. Therefore, we have

$$\frac{1}{g^2(\mu_1)} - \frac{1}{g^2(\mu_2)} = -\frac{19}{6(4\pi)^2} \log \frac{\mu_2^2}{\mu_1^2}. \quad (1.2)$$

For $\mu_1 > \mu_2$, we have $g^2(\mu_1) < g^2(\mu_2)$, and hence the theory is asymptotically free.

- ii) From the one-loop relation in Sheet 2, Problem 5, we have

$$\Lambda_{\text{weak}} = (100 \text{ GeV}) \exp\left(-\frac{6\pi}{10} \cdot 30\right) = 2.76 \times 10^{-14} \text{ eV}. \quad (1.3)$$

- iii) Even if the $SU(2)$ symmetry is not broken by the Higgs, the condensate still breaks the $SU(2)$.

- iv) For $SU(3)$, $N_c = 3, N_f = 6, N_s = 0$. Then, using the one-loop relation in Sheet 2, Problem 5, we find

$$\Lambda_{\text{QCD}} = (100 \text{ GeV}) \exp\left(-\frac{6\pi}{21} \cdot 10\right) = 1.26 \times 10^{-2} \text{ GeV}. \quad (1.4)$$

When $\alpha_W = \alpha_s$, we find

$$\mu = \left(\frac{\Lambda_{\text{weak}}^{10}}{\Lambda_{\text{QCD}}^{21}}\right)^{-1/11} = 7.7 \times 10^{16} \text{ GeV}. \quad (1.5)$$

b) For $U(1)$ coupled to massless Weyl fermions of charges Y_i and scalars of charges $\tilde{Y}_i$, use

$$\frac{1}{g^2(\mu)} = \frac{1}{g_0^2} + \frac{2 \sum_i Y_i^2 + \sum_i \tilde{Y}_i^2}{3(4\pi)^2} \log \frac{\Lambda_{UV}^2}{\mu^2}. \quad (1.6)$$

Decide whether hypercharge is asymptotically free. With $\alpha_Y(100 \text{ GeV}) \simeq 1/98$, find the scales where $\alpha_s = \alpha_Y$ and $\alpha_W = \alpha_Y$.

Solution. Since this is always positive, the hypercharge is not asymptotically free. In particular, for one generation

$$\sum_{\text{Weyl}} Y_i^2 = 6 \left(\frac{1}{6}\right)^2 + 3 \left(\frac{2}{3}\right)^2 + 3 \left(-\frac{1}{3}\right)^2 + 2 \left(-\frac{1}{2}\right)^2 + (-1)^2 = \frac{10}{3}. \quad (1.7)$$

For the scalars, we only have the Higgs

$$\sum_{\text{scalars}} \tilde{Y}_i^2 = 2 \left(\frac{1}{2}\right)^2 = \frac{1}{2}. \quad (1.8)$$

Therefore, we have

$$2 \sum_i Y_i^2 + \sum_i \tilde{Y}_i^2 = 20 + \frac{1}{2} = \frac{41}{2} \quad (1.9)$$

Hence, through the one-loop relation in Sheet 2, Problem 5, we find

$$\Lambda_Y = 100 \text{ GeV} \exp\left(\frac{1176\pi}{41}\right) = 1.36 \times 10^{41} \text{ GeV}. \quad (1.10)$$

Following the same procedure as in the previous sections, we find

$$\mu_{sY} = (\Lambda_{QCD}^{42} \Lambda_Y^{41})^{1/83} = 2.28 \times 10^{19} \text{ GeV}, \quad (1.11)$$

and

$$\mu_{WY} = (\Lambda_{\text{weak}}^{19} \Lambda_Y^{41})^{1/60} = 3.59 \times 10^{20} \text{ GeV}. \quad (1.12)$$

2. Sheet 4, Problem 2: Charge of a New Multiplet

A hypothetical particle transforms as a **3** of $SU(2)$ and a **6** of $SU(3)$, with hypercharge Y . Find its electric charges after electroweak symmetry breaking.

Solution. The charge is defined as $Q = T^3 + Y$. **3** corresponds to $j = 1$, and hence T^3 has three eigenvalues, namely $-1, 0, +1$. Therefore, this particular particle contains three components with charges $Y - 1, Y, Y + 1$.

3. Sheet 4, Problem 4: Counting Flavour Parameters

With N generations, count the physical parameters in the quark Yukawa matrices and classify them as masses, mixing angles, and complex phases. Then assume neutrino masses arise from a dimension-five operator and perform the corresponding count for the charged-lepton Yukawa matrix together with that operator.

Solution. The two Yukawa matrices, y^u, y^d , have $2N^2$ complex parameters, and hence $2N^2$ real parameters and $2N^2$ phase angles. Since the kinetic term is invariant under $U(N)$ transformation, the

physics remains under such transformations as long as the Yukawa matrices transform as

$$y^u \rightarrow V^\dagger y^u U^u, \quad Y^d \rightarrow V^\dagger y^d U^d, \quad (3.1)$$

where $Y, U^u, U^d \in U(N)$. $U(N)$ can be thought of as a decomposition of $O(N)$ and $U(1)^n$. $\dim U(n) - \dim O(n) = N^2 - \frac{1}{2}N(N-1) = \frac{1}{2}N(N+1) = n$. Therefore, there are

$$2N^2 - \frac{3}{2}N(N-1) = \frac{1}{2}N^2 + \frac{3}{2}N \quad (3.2)$$

real physical parameters, whereas $2N$ of them are physical mass parameters, and the rest are the mixing angles. Meanwhile, if all of Y, U^u, U^d shifts by the same phase, the Yukawa potential remains unchanged, we must take away an $U(1)$ phase. Hence, there are

$$2N^2 - \frac{3}{2}N(N+1) + 1 = \frac{1}{2}N^2 - \frac{3}{2}N + 1 \quad (3.3)$$

physical complex phases.

A similar analysis works in the electron-neutrino sector. The mass Lagrangian is given by

$$-\bar{L} y^e e_R H + \frac{1}{2\Lambda} \kappa_{ij} (L_i H) (L_j H) + h.c. \quad (3.4)$$

Here y^e is an arbitrary complex $N \times N$ matrix, while κ is a complex $N \times N$ symmetric matrix. Therefore, there are N^2 real parameters and N^2 complex phases in the electron Yukawa coupling and $\frac{1}{2}N(N+1)$ real parameters and $\frac{1}{2}N(N+1)$ complex phases in the dimension 5 operator. The Kinetic Lagrangian has a $U(N)_L \times U(N)_e$ symmetry, which the couplings transforms as

$$y^e \rightarrow W^\dagger y^e U^e, \quad \kappa \rightarrow W^T \kappa W. \quad (3.5)$$

Note that there is no overall $U(1)$ phase that preserves the dimension 5 coupling. Therefore, we have $N(N+3)/2$ real physical parameters and $N(N-1)/2$ physical phases. These are N lepton masses, N neutrino masses, $N(N-1)/2$ mixing angles and $N(N-1)/2$ complex phases.

4. Sheet 4, Problem 5*: A Two-Generation Cabibbo Angle

For two quark families, let

$$\mathcal{L}_{\text{mass}} = -m_{ij} \bar{q}^i q^j - \tilde{m}_{ij} \bar{\tilde{q}}^i \tilde{q}^j, \quad m = \begin{pmatrix} 0 & a \\ a^* & b \end{pmatrix}, \quad \tilde{m} = \begin{pmatrix} 0 & c \\ c^* & d \end{pmatrix}, \quad (4.1)$$

with $a, c \in \mathbb{C}$ and $b, d \in \mathbb{R}$.

- i) Rephase quarks to make a, c real and positive.
- ii) Let rotations $R(\theta), R(\tilde{\theta})$ diagonalise the two matrices into $(m_1, -m_2)$ and $(\tilde{m}_1, -\tilde{m}_2)$. Show

$$\tan \theta = \sqrt{\frac{m_1}{m_2}}, \quad \tan \tilde{\theta} = \sqrt{\frac{\tilde{m}_1}{\tilde{m}_2}}. \quad (4.2)$$

- iii) Derive $\theta_C = \theta - \tilde{\theta}$ from the weak charged-current mixing matrix.

Solution.

- i) m can be written as

$$m = \begin{pmatrix} 0 & |a|e^{i\alpha} \\ |a|e^{-i\alpha} & b \end{pmatrix}. \quad (4.3)$$

The kinetic term is invariant under

$$q^1 \rightarrow e^{-i\alpha} q^1, \quad (4.4)$$

whereas m transforms as

$$m_{12} \rightarrow e^{i\alpha} m_{12}, \quad m_{21} \rightarrow e^{-i\alpha} m_{21}. \quad (4.5)$$

Therefore, we can always make $a, c \in \mathbb{R}^+$.

ii) We have

$$m = R(\theta)^{-1} \text{diag}(m_1, -m_2) R(\theta). \quad (4.6)$$

Note that the top right entry is zero, we must have

$$m_1 \cos^2 \theta - m_2 \sin^2 \theta = 0 \implies \tan \theta = \sqrt{\frac{m_1}{m_2}}, \quad (4.7)$$

and similarly for $\tilde{\theta}$.

iii)

$$R(\theta_c) = R^{-1}(\tilde{\theta}) R(\theta) \implies \theta_c = \theta - \tilde{\theta}. \quad (4.8)$$

5. Sheet 4, Problem 6*: A Second Higgs Doublet

Alongside the Standard Model Higgs $H : (\mathbf{2}, \mathbf{1})_{+1/2}$, introduce $\phi : (\mathbf{2}, \mathbf{1})_{-1/2}$, with the entries indicating $SU(2) \times SU(3)$ representations and $U(1)_Y$ charge.

- i) Find the electric charges of the two components of ϕ .
- ii) Construct the most general renormalisable gauge-invariant potential of H and ϕ and count its independent real parameters.
- iii) Suppose the vacuum is

$$\langle H \rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ v \end{pmatrix}, \quad \langle \phi \rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} u + iw \\ 0 \end{pmatrix}, \quad u, v, w \in \mathbb{R}. \quad (5.1)$$

Does $\langle \phi \rangle$ break electromagnetism? Find the masses of $W_\mu^\pm = (W_\mu^1 \mp iW_\mu^2)/\sqrt{2}$ and of the appropriate mixtures of W_μ^3, B_μ . Identify the charges of the scalar modes remaining after symmetry breaking.

- iv) List all permitted Yukawa couplings of H and ϕ to the Standard Model fermions.

Solution.

- i) For $\mathbf{2}$ of $SU(2)$, T^3 has eigenvalues $\pm \frac{1}{2}$. Hence, the EM charges are $-1, 0$.
- ii) Since $[H] = [\phi] = 1$, we can at most have an order 4 coupling. Meanwhile, we must have two scalar fields coming in pair as well. Finally, to respect $U(1)$ gauge symmetry, the only possible interactions we are left with are

$$\begin{aligned} V = & m_1^2 H^\dagger H + m_2^2 \phi^\dagger \phi - [m_{12}^2 H^T i\sigma^2 \phi + \text{h.c.}] \\ & + \lambda_1 (H^\dagger H)^2 + \lambda_2 (\phi^\dagger \phi)^2 + \lambda_3 (H^\dagger H)(\phi^\dagger \phi) + \lambda_4 |H^T i\sigma^2 \phi|^2 \\ & + [\lambda_5 (H^T i\sigma^2 \phi)^2 + \lambda_6 (H^\dagger H)(H^T i\sigma^2 \phi) + \lambda_7 (\phi^\dagger \phi)(H^T i\sigma^2 \phi) + \text{h.c.}], \end{aligned} \quad (5.2)$$

Therefore, there are 6 independent real parameters.

- iii) In the vacuum, we have

$$Q\langle \phi \rangle = (T^3 + Y)\langle \phi \rangle = \begin{pmatrix} 0 & 0 \\ 0 & -1 \end{pmatrix} \langle \phi \rangle = 0. \quad (5.3)$$

Therefore, the $U(1)_{\text{EM}}$ symmetry survives.

Now, expanding the terms quadratic in the gauge fields around the vacuum, we find

$$\mathcal{L}_{\text{mass}} = \frac{v^2 + u^2 + w^2}{8} [g^2 ((W_\mu^1)^2 + (W_\mu^2)^2) + (gW_\mu^3 - g'B_\mu)^2]. \quad (5.4)$$

Therefore,

$$m_W^2 = \frac{g^2}{4}(v^2 + u^2 + w^2). \quad (5.5)$$

Defining

$$Z_\mu = \cos \theta_W W_\mu^3 - \sin \theta_W B_\mu, \quad A_\mu = \sin \theta_W W_\mu^3 + \cos \theta_W B_\mu, \quad (5.6)$$

where

$$\cos \theta_W = \frac{g}{\sqrt{g^2 + g'^2}}, \quad (5.7)$$

we obtain

$$m_Z^2 = \frac{g^2 + g'^2}{4}(v^2 + u^2 + w^2), \quad m_\gamma^2 = 0. \quad (5.8)$$

iv) Since ϕ has hypercharge $Y_\phi = -1/2$, the allowed interactions are

$$\mathcal{L}_{\phi f} = -\bar{Q}_L y_\phi^u \phi u_R - \bar{Q}_L y_\phi^d \tilde{\phi} d_R - \bar{L} y_\phi^e \tilde{\phi} e_R + \text{h.c.}, \quad (5.9)$$

where

$$\tilde{\phi} = i\sigma^2 \phi^*. \quad (5.10)$$